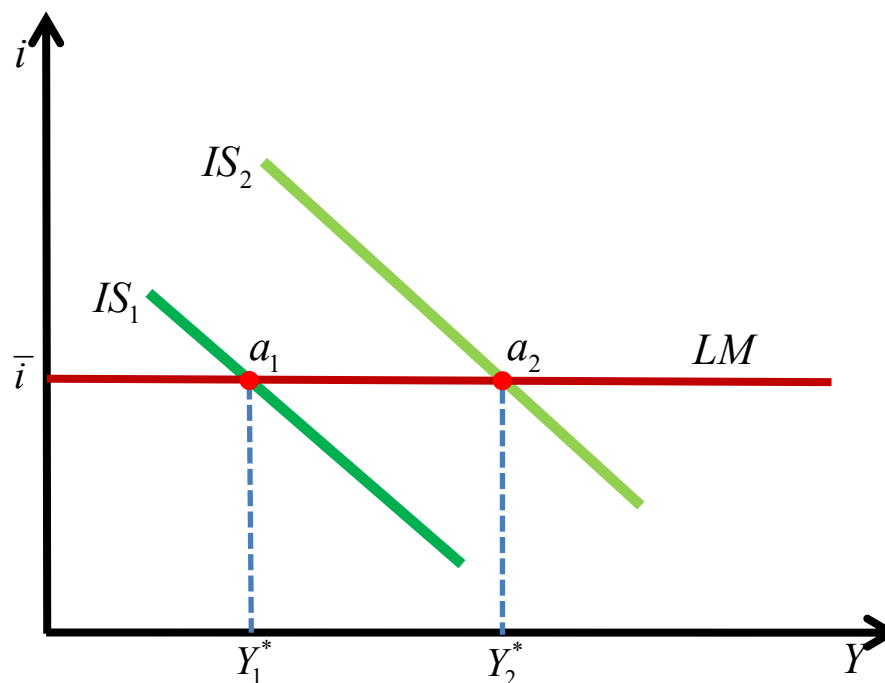


Question 1: The macroeconomic system of Acheron is described by a standard IS-LM model. Denote by “ $a_1$ ” the initial equilibrium. For unknown reasons, the local consumers suddenly increase their autonomous level of consumption ( $c_0$ ). Denote by “ $a_2$ ” the new equilibrium. Which of the following is the appropriate graph? Choose one and motivate your answer. Explain the effects of this change in  $c_0$  on the output level ( $Y$ ) and its composition.



*Answer: Local consumers increase their autonomous level of consumption ( $c_0$ ). This affects the vertical intercept of the IS curve. The curve shifts to the right. Output in equilibrium increases. Consumption and investment both increase.*

Question 2: Are the following statements true or false? Motivate your answer.

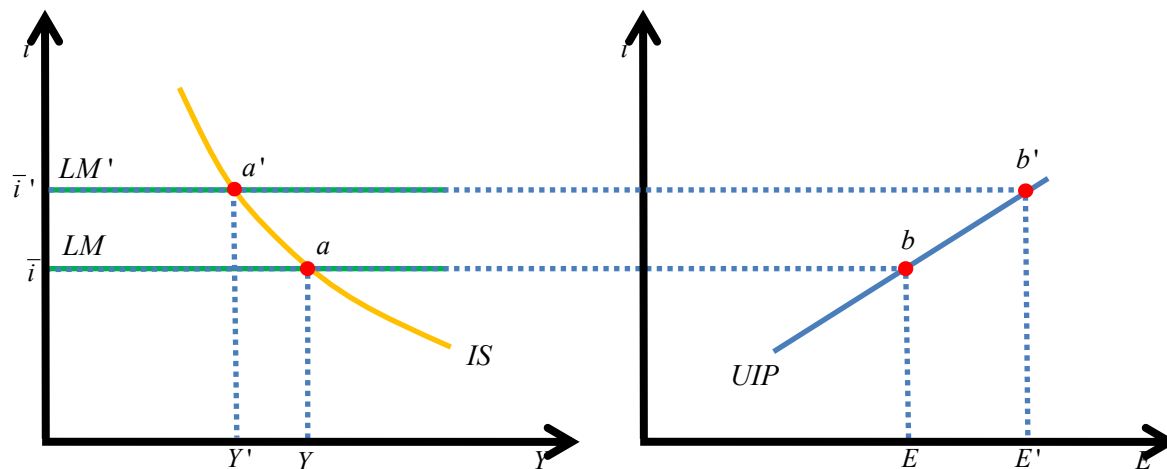
- a) The standard Solow model predicts that, following a sufficiently large decrease in the depreciation rate, the stock of capital per worker will converge to a strictly positive value in the long run.

*True. Since the growth rate will converge to zero,  $K/N$  will converge to a positive value.*

- a) The standard Solow growth model predicts that countries starting from a lower level of GDP per worker do necessarily grow faster than countries starting from a higher level.

*False. This is a possibility not a necessity. What matters is the distance from the country's steady state.*

Question 3: Krypton trades freely goods and services with the rest of the Empire, operating under a flexible exchange rate regime with perfect capital mobility, and its economy system can be described by the standard open-economy IS-LM model. Suppose that both the domestic and foreign price levels are fixed (so that  $\pi = \pi^e = 0$ ), and that the Marshall-Lerner condition is satisfied. The Central Bank implements a contractionary change in its monetary policy, increasing  $i$ . Which of the following is the appropriate graph? Choose one and motivate your answer.



*Answer: The domestic Central Bank increases  $i$ . The LM shifts upwards. Suddenly, foreign bonds become less attractive for a given domestic interest rate. The UIP implies that  $E$  has to rise on impact in order to be able to depreciate over time and compensate for the interest differential. Output drops as we move along the IS curve. Consumption and investment drop (because of higher  $i$  and lower  $Y$ ), and the sudden appreciation of  $E$  influences negatively  $NX$ .*

Question 4: Are the following statements true or false? Motivate your answer.

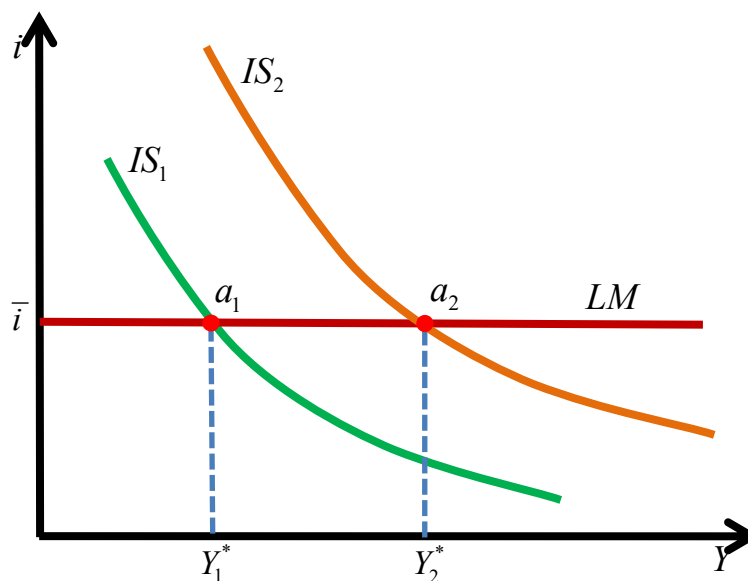
- a) Bank A grants €20 worth of mortgage loans to Customer A, crediting the principal on his account, and sells €30 worth of securities to customer B, charging the amount to her account. These transactions create €10 worth of additional broad money.

*False. The transaction destroys €10 worth of broad money.*

- a) Customer A uses the loan's principal to buy a new car, paying with a check; the car dealer deposits the check in Bank B. These operations decrease the supply of broad money by €30 and decrease Bank A's central bank reserves by €20.

*False. The transaction does not affect broad money.*

Question 5: The macroeconomic system of Dagobah is described by a standard IS-LM model. Denote by “ $a_1$ ” the initial equilibrium. For unknown reasons, the local Governor suddenly increases both military expenditure - which is part of government spending ( $G$ ) - and taxes ( $T$ ), by the **same** amount. Denote by “ $a_2$ ” the new equilibrium. Which of the following is the appropriate graph? Choose one and motivate your answer. Explain the effect of these changes on the output level ( $Y$ ) and its composition.



*Answer: The local Governor suddenly increases both government spending ( $G$ ) and taxes ( $T$ ), by the **same** amount, so that  $\Delta G = \Delta T > 0$ . Being  $c_1 < 1$ ,  $c_1 |\Delta T| < |\Delta G|$ . Hence, aggregate demand will rise. This shifts the IS curve to the right. Evidently:  $Y \uparrow, I \uparrow, C \uparrow$ .*

Question 6: Are the following statements true or false? Motivate your answer.

- a) Quantitative easing does not affect the supply of broad money.

*False. QE does affect the supply of broad money, as any transaction that buys or sell financial assets from/to the non-financial private sector.*

- a) Under inflation targeting, if the central bank is missing its target, then the actual unemployment rate remains necessarily above the natural rate.

*False. Not necessarily. This depends on many factors and in particular on how expectations are formed.*